INTRODUCTION TO HYBRID AND ELECTRIC VEHICLES

Subject Code : 23A0360Se

UNIT V

Electric and Hybrid Vehicle Design

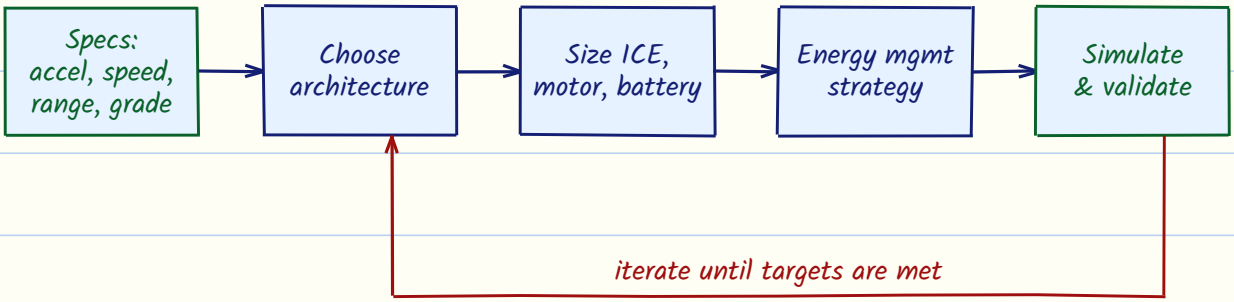
I. Introduction to Hybrid Vehicle Design

Designing a hybrid / electric vehicle means choosing the **architecture** and then **sizing every component** (engine, motor, power electronics, battery, drive train) so that the vehicle meets performance targets with the lowest fuel use, cost and weight.

Design requirements (specifications)

- **Acceleration**: 0-100 km/h in t_a seconds (e.g. 10 s). **Top speed** (e.g. 150 km/h). **Gradeability**: climb x % slope at given speed (e.g. 20 % at low speed, 5 % at 100 km/h).
- **Range** / fuel economy on standard drive cycles (**NEDC, WLTP, FTP-75, Indian IDC**), cruise range, emission limits.
- Weight, cost, package space, safety (high-voltage), life 8-10 years, comfort (NVH) and charging time.

Design procedure



- **Step 1** fix vehicle parameters (m, C_D, A_f, μ_{rr}, wheel radius); **Step 2** compute power demand for each requirement; **Step 3** size motor and ICE; **Step 4** size battery for power and energy; **Step 5** develop control strategy; **Step 6** simulate on drive cycles (MATLAB/Simulink, ADVISOR, Autonomie) and optimise.

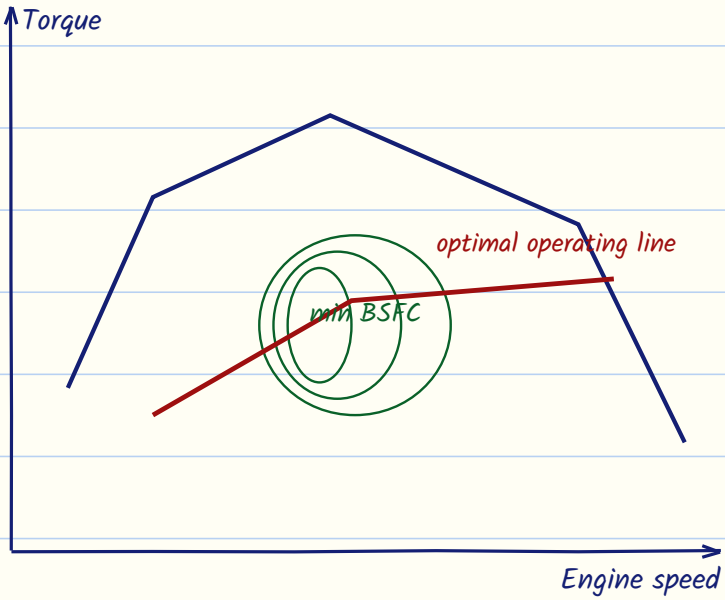
Degree of hybridisation

$$H = P_{em} / (P_{em} + P_{ICE})$$

$H = 0$ conventional car, $H = 1$ pure EV. Micro / mild hybrid: $H \approx 0.1$–0.2; full hybrid: 0.3–0.5; plug-in / EREV: 0.5–0.8.

2. Matching the Electric Machine and the ICE

The two power sources must complement each other: the **ICE** is good for long-duration, steady power at high load (but inefficient at low load); the **electric machine** is good for transients, low speed torque and regeneration.



Matching ICE and motor:
Operate ICE on the optimal line (best BSFC contours). Motor fills the gap between demand and ICE output and absorbs excess power.

Parallel hybrid – torque coupling

$$T_{out} = k_1 T_{ICE} + k_2 T_{EM}$$

- Torque coupler (gear / belt) adds torques; speeds are proportional: $\omega_{out} = \omega_{ICE}/k_1 = \omega_{EM}/k_2$. Used in Honda IMA, parallel hybrids. Motor and engine ratings chosen so that **ICE alone** satisfies cruising and mild gradients, motor adds in peak demands.

Speed coupling (series-parallel / power split)

$$\omega_{out} = k_1 \omega_{ICE} + k_2 \omega_{EM}$$

- Planetary gear adds speeds while torques are proportional, so the engine speed can be varied independently of the wheel speed (CVT action).

Matching rules

- **ICE power** is selected to meet steady-state demand: $P_{ICE} \geq P_{cruise\ at\ v_\{max\}}$ (or at max gradient and speed) / η_t, usually 50-70 % of the equivalent conventional engine.
- **Series hybrid**: engine-generator rating = average power demand (about 25-30 kW for a car) + battery charging power; motor must meet the **peak** demand alone.
- Operate ICE on the **optimal operating line** on the BSFC map (minimum fuel for each power). Choose gear ratios so that cruising lies near the line.
- Match maximum speeds: ICE max rpm / gear ratio ≥ vehicle top speed; motor base speed ≈ engine idle-to-mid-speed range.

3. Sizing of the Propulsion Motor

The motor must provide peak power for **acceleration**, continuous power for **top-speed cruising** and enough torque for **gradeability**. The rated power is the maximum of the three.

(a) Power for acceleration (0 to v_f in t_a s)

$$P_m = (\delta\, m / 2\, t_a)\,(v_f^2 + v_b^2) + (2/3)\, m\, g\, \mu_{rr}\, v_f + (1/5)\, \rho\, C_D\, A_f\, v_f$$

v_b = base speed (vehicle speed at motor base speed); v_f = final speed; δ = rotational mass factor (1.05).

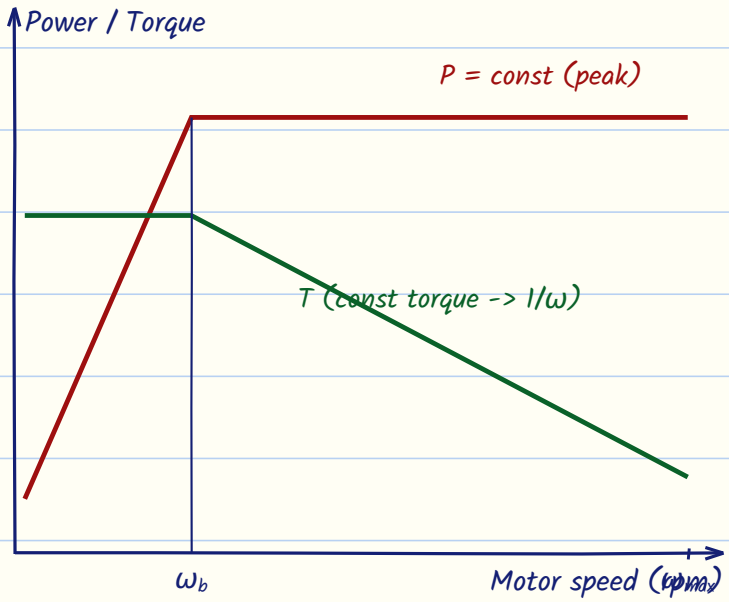
- Example: $m = 1500$ kg, $v_f = 100$ km/h = 27.8 m/s, $v_b = 50$ km/h = 13.9 m/s, $t_a = 10$ s, $\mu_{rr} = 0.012$, $\rho C_D A_f = 0.792$.
- Term 1 = $(1.05 \times 1500 / 20)(27.8^2 + 13.9^2) = 78.75 \times 966 \approx$ **76 kW**; Term 2 = $(2/3)(1500)(9.81)(0.012)(27.8) \approx 3.3$ kW; Term 3 = $0.2 \times 0.792 \times 27.8^3 \approx 3.4$ kW.
- **$P_m \approx 83$ kW** (peak). Continuous rating for steady 100-130 km/h cruise is much lower (15-25 kW); the motor is rated for peak with short-time overload (1.5-2×).

(b) Power for top speed and gradient

$$P_m = (v / \eta_t) [m g \mu_{rr} \cos\alpha + m g \sin\alpha + \tfrac{1}{2} \rho C_D A_f v^2]$$

(c) Torque and speed rating

- Base speed ω_b: motor rated torque $T_{rated} = P_{rated} / \omega_b$. Larger $x = \omega_{max}/\omega_b$ gives smaller T_{rated} for the same peak power (smaller motor and inverter) but demands good field weakening.
- Gear ratio from top speed: $G = \omega_{max} r / v_{max}$. Example: 9000 rpm (942 rad/s), $r = 0.3$ m, $v_{max} = 150$ km/h = 41.7 m/s -> $G = 942 \times 0.3 / 41.7 =$ **6.8 ≈ 7**.



Rated power from:
(1) acceleration time
(2) top speed (cruise)
(3) max gradient at speed
Take the LARGEST value.
Constant-power ratio $x = \omega_{max}/\omega_b$
typ. 3-4 (IM / IPM).

Remember: acceleration time normally decides the peak power; gradeability decides the maximum torque; top speed decides the continuous power.

4. Sizing of Power Electronics

- **Inverter current rating** from motor peak power: $I_{rms} = P_{peak} / (\sqrt{3}\, V_{LL} \cos\varphi\, \eta)$. Example: 83 kW, V_{LL} = 200 V, pf = 0.9, η = 0.95 -> I_{rms} = 83000/(1.732×200×0.9×0.95) = **280 A rms (about 396 A peak)**.
- **DC-link voltage** $V_{dc} \geq \sqrt{2}\, V_{LL}/m_a$ (e.g. 200 V line gives about 300-350 V bus) plus margin 20-30 % for regenerative overshoot; semiconductor rating = 1.5-2 × V_{dc} (e.g. 650 / 1200 V IGBT).
- **Switching device**: IGBT (high power), MOSFET (low voltage, <200 V), **SiC MOSFET** (high efficiency, high

frequency). Choose switching frequency 8-20 kHz; select gate driver and snubber.

- **DC-link capacitor** to absorb ripple current (film capacitor), sized for ripple $I_{c,rms}$ and voltage ripple.
- **Thermal design**: loss P_{loss} = conduction + switching; heat-sink / liquid cooling to keep junction below 125-150 °C; DC-DC converter sized for auxiliary load (2-3 kW, 12 V).
- Include protection: over-current, over-voltage, short circuit, temperature, isolation monitoring.

5. Sizing of the Drive System

- **Transmission**: single-speed reduction (G = 6-10) for EVs; two-speed for high-performance or heavy EV; CVT / planetary in HEVs. Efficiency of gear 95-98 %.
- Check wheel torque: $T_{wheel} = T_m\, G\, \eta_t$; tractive force $F = T_{wheel}/r$ must exceed required F_{te} at start, and be below adhesion limit $\mu_a N$.
- Select final drive and differential to meet acceleration, gradeability and maximum speed together; check motor operating on efficiency map for the drive cycle.
- Hybrids: choose gear ratio between engine and wheels so that engine rpm at cruise lies near the optimum BSFC

region.

6. Selection of Energy Storage Technology

Energy storage options: **Li-ion / NiMH batteries**, **ultracapacitors**, **flywheels**, **fuel cell** (source), or a **combination** (hybrid storage). Selection depends on required **energy** (range) and **power** (acceleration / regen).

Device	Specific energy	Specific power	Life	Suitable for
Li-ion	100-250 Wh/kg	300-1500 W/kg	1000-3000 cyc	BEV, PHEV, HEV
NiMH	60-120 Wh/kg	200-1000 W/kg	500-1500 cyc	HEV
Ultracapacitor	3-10 Wh/kg	5000+ W/kg	>500 000 cyc	Peak power, regen
Flywheel	10-50 Wh/kg	~5000 W/kg	Very long	Buses, special
Fuel cell + H_2	High (system)	Low-medium	5000 h+	FCEV, range

Sizing of the battery pack

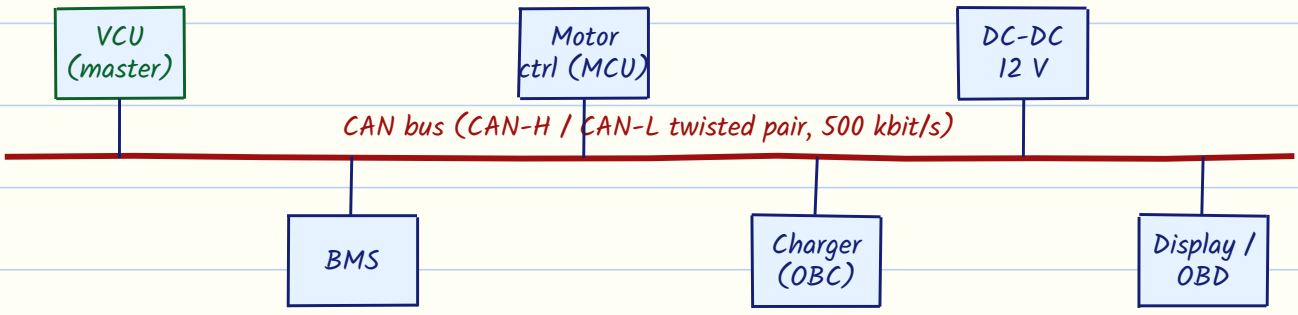
- **Energy**: E_{pack} = (range × Wh/km) / usable DoD. Example: 150 km × 150 Wh/km = 22.5 kWh; with 80 % usable window -> **28 kWh** pack.

- **Power**: $P_{batt,max} \geq P_{motor,peak} / (\eta_m \eta_c)$ and also ≥ regen power limit. For HEVs power, not energy, sizes the battery (1-2 kWh with high C-rate); for EVs energy dominates.
- **Series-parallel arrangement**: $N_s = V_{pack}/V_{cell}$; capacity Ah = E_{pack}/V_{pack}. Example: 360 V / 3.6 V = 100 cells in series; 28 kWh / 360 V = 78 Ah -> 78 Ah cells (or e.g. 2 parallel strings of 39 Ah). Mass = 28 000 / 150 Wh/kg ≈ **187 kg**.
- Add BMS, cooling, contactors, fuse; check C-rate (P / E) and cycle life; consider **hybrid storage** (battery + ultracap) to reduce peak stress and extend battery life.

7. Communications and Supporting Subsystems

Communications

In-vehicle communication network



- Many ECUs must share data: **VCU** (vehicle control unit) gets pedal, brake, gear and SOC signals and sends torque commands to the motor controller and engine ECU; BMS

reports voltage, SOC, temperature; charger and DC-DC report status.

- Protocols: **CAN (ISO 11898)** most common, **LIN** (low-cost body control), **FlexRay** (fast, safety-critical), **Ethernet** for infotainment, **PLC / ISO 15118** and CHAdeMO / CCS for charger communication.

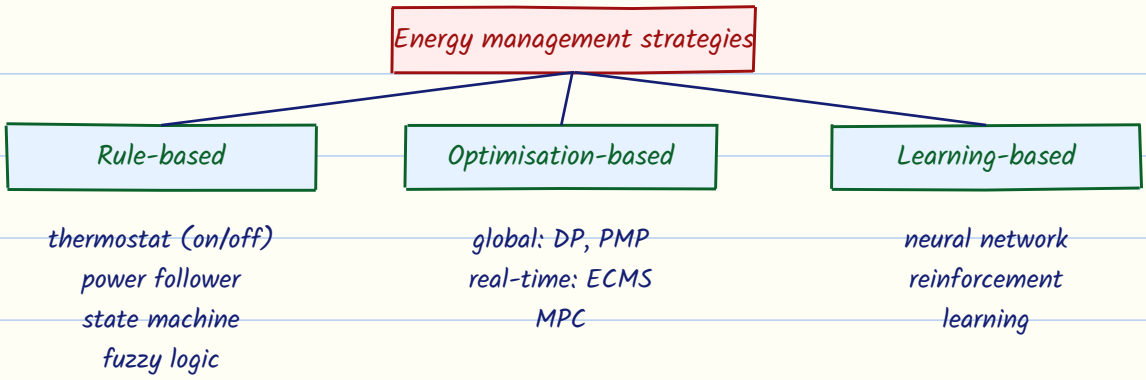
Supporting subsystems

- **Auxiliary power supply**: DC-DC converter (high voltage -> 12 V) for lamps, ECUs, wipers; 12 V battery retained for safety.
- **Thermal management**: separate cooling loops (liquid) for battery, motor and inverter; PTC / heat-pump for cabin heating; battery heater for cold climates.
- **Brake system**: **regenerative braking blending** with hydraulic brakes; electric vacuum pump for brake booster; electric power steering (EPS) and electric A/C compressor.
- **Safety**: high-voltage interlock loop (HVIL), isolation monitoring, main contactors with pre-charge, crash disconnect, emergency service plug, fuse; on-board charger and charging port.

8. Energy Management Strategies (EMS) in Hybrid and Electric Vehicles

The **EMS** (supervisory controller) decides at every instant the power split between ICE, motor and battery to minimise fuel (or energy) consumption and emissions while keeping the battery SOC in the safe window and satisfying driver demand. In pure EVs the EMS manages battery / ultracap / fuel-cell sharing and regeneration.

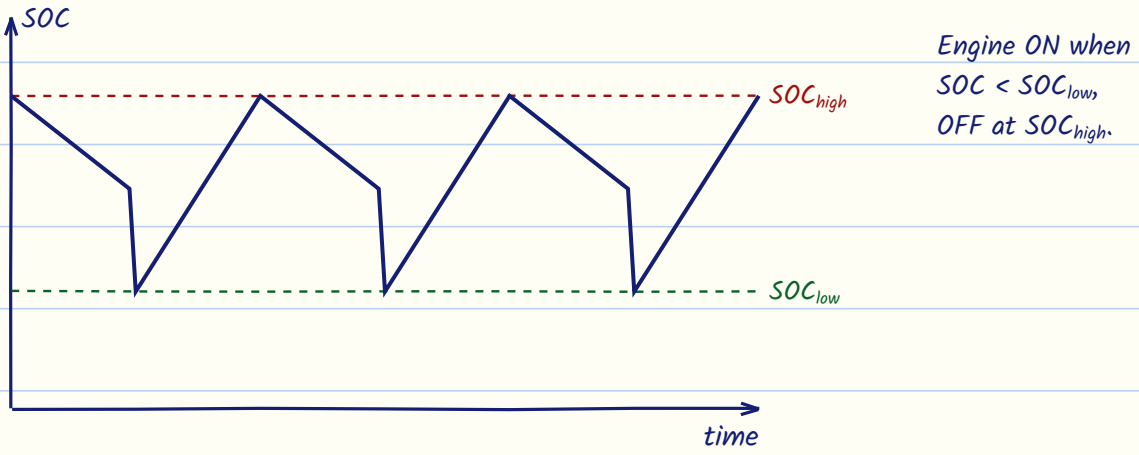
(a) Classification



- **Rule-based (heuristic)**: predefined rules from engineering experience; very simple, real-time, but not optimal.
  - **Thermostat (on-off) control** - engine switched on when SOC < SOC_{low}, run at best-efficiency point until SOC_{high}, then off. Used in series hybrids.
  - **Power follower** - engine follows demand; battery supplies the remaining; **modified** versions keep engine near

optimum line. **State machine / fuzzy logic** handle nonlinearity.

- **Optimisation-based**: minimise a cost function $J = \int (\dot{m}_f + \lambda(SOC)\cdot\dots) \, dt$.
  - **Global**: dynamic programming (DP), Pontryagin's minimum principle (PMP) - optimal but need the whole drive cycle in advance (offline, benchmark).
  - **Real-time**: **ECMS** (equivalent consumption minimisation strategy), model predictive control (MPC), adaptive ECMS.
- **Learning-based / predictive**: neural networks, reinforcement learning, GPS / traffic-based predictive control (preview information).



$$\text{ECMS: } \dot{m}_{eq}(t) = \dot{m}_f(t) + s(t)\, P_{batt}(t) / Q_{LHV}$$

$s(t)$ = equivalence factor converting electric power into equivalent fuel (Q_{LHV} = lower heating value of fuel, 42-44

MJ/kg). At each instant the split minimising $\dot{m}_{eq}$ is chosen.

(b) Comparison

Strategy	Optimality	Computation	Needs trip info	Real-time use
Rule-based	Low-medium	Very low	No	Yes - production cars
DP / PMP	Global optimum	Very high	Yes (full cycle)	No (benchmark)
ECMS	Near optimal	Low-medium	No / partial	Yes
MPC	Near optimal	Medium-high	Preview	Yes
Learning	Good, adaptive	Training heavy	Optional	Yes (after training)

(c) Implementation

- Run on the VCU (microcontroller) as periodic task (10-100 ms). **Inputs**: driver pedal demand, vehicle speed, SOC, battery temperature, engine speed and temperature, road / trip data. **Outputs**: torque commands to engine and motor, mode (EV / hybrid / regen), battery charge limit.
- Rules or look-up tables (offline-optimised maps) are

stored in memory; **SOC feedback** keeps charge sustaining (adjust s(t) or thresholds).

- Development path: model-in-the-loop (MIL, Simulink) -> software-in-the-loop (SIL) -> **hardware-in-the-loop (HIL)** -> vehicle testing on chassis dynamometer; auto-code generation (Embedded Coder / dSPACE), then calibration.
- Add constraints: drivability (no torque jerk during engine start), emissions / catalyst temperature, battery life (limit C-rate and DoD), safety fallback (limp-home mode).

Important questions: (1) Matching of ICE and electric machine. (2) Sizing of propulsion motor with example. (3) Sizing of battery and power electronics. (4) Communication and supporting subsystems. (5) Classify and compare EMS strategies (ECMS, rule-based, DP).